

ICACHE

Design Specification

Ver 0.1

DM270

Department: IMAGING

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HISTORY

Version	Date	Author	Notes
Ver beta.0	03/01/02	tos	Initial Release for DM270
Ver beta.1	03/05/02	tos	some in TBD are fixed
Ver 0.0	04/04/02	tos	Ica_req4word
Ver 0.1	06/17/02	snata	Update tag structure. Update Normal Function state machine. Add mtc_sdsl and clk_tag signal. Update register offset address. Add Test mode register.

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1. ABSTRACT

DM270 has ARM ICACHE (Instruction Cache), to reduce accesses to an external memory, which takes more cycles than internal memory.

2. REFERENCES

ARM7TDMI datasheet ARM DDI 0029E, Aug. 1996, ARM Ltd.

3. GLOSSARY

ICACHE	Instruction Cache
ARM	ARM7TDMIE
MTC	Memory Traffic Controller
BUSC	Bus Controller

4. FUNCTION

ICACHE contains 8kbyte RAM for caching instruction. ICACHE selects only instruction mapped external memory(0x100000-0x80FFFFFF) by using nOPC signal from ARM. ICACHE feature is below.

- ◆ 2way-set-associative instruction cache
- ◆ 4-word (128bit) line size
- ◆ Total 512-line (256-line x 2way)
- ◆ Update data by LRU(Least Recently Used)
- ◆ Check tag-RAM according to address from ARM and execute below action in one wait cycle.
 - ◆ When HIT, ARM reads the data stored in RAM in ICACHE.
 - ◆ When MISS, the communication between ARM and MTC starts.
- ◆ Use BRY type RAM (BRY01024032080 x 2) to store the data
- ◆ One valid bit per 2byte data in tag-RAM
- ◆ No Dirty bit in tag-RAM (no support write)
- ◆ Support ICACHE disable(bypass) mode
- ◆ Selectable 1word/4word access (between ICACHE and MTC)

4.3 Tag-RAM

44	43:24	23:22	21:2	1:0
LRU	Way0 Memory Address[23:12]	Way0 Vbit[7:0]	Way1 Memory Address[23:12]	Way1 Vbit[7:0]

Figure 3 : Structure of tag-RAM

4.4 MTC I/F

MTCからのデータはICAMODEレジスタのREQ4Wの値に応じてICACHEに渡されます。SDRAMへのアクセスで、REQ4Wが1にセットされている場合には前後4ワードを一度に取り込みます。

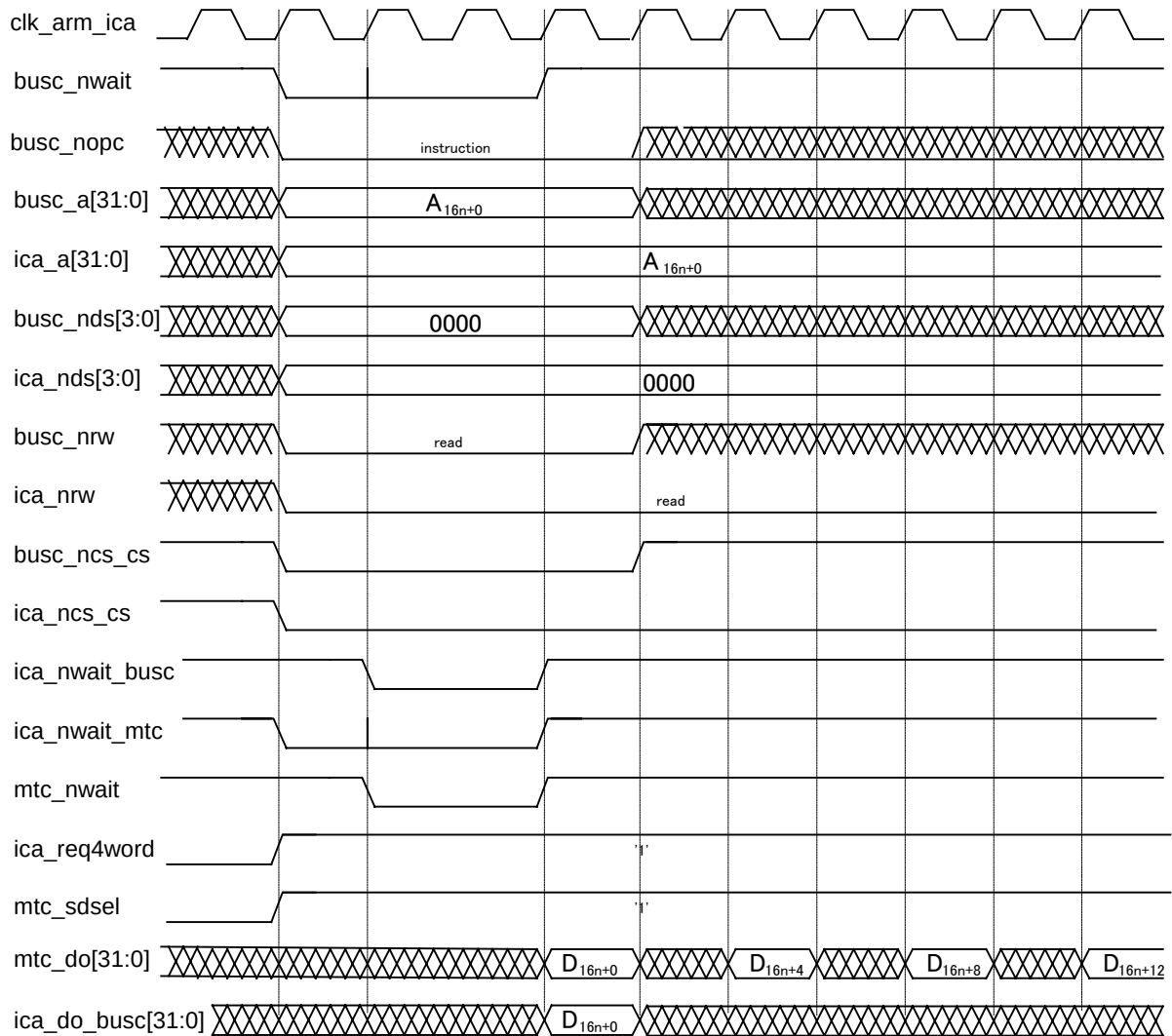


Figure 4 : MTC I/F timing (req4word = 1)

ARMから要求されたアドレスが、 $16n+0$ ($n=0,1,2,\dots$) で表現される場合は、続く3word分のデータがARMを開放した後にMTCから送られてキャッシュされます。ARM要求アドレスが $16n+4/16n+8/16n+12$ で表現される場合は上図のbusc_a[31:0]、mtc_do[31:0]は以下で読み替えてください。

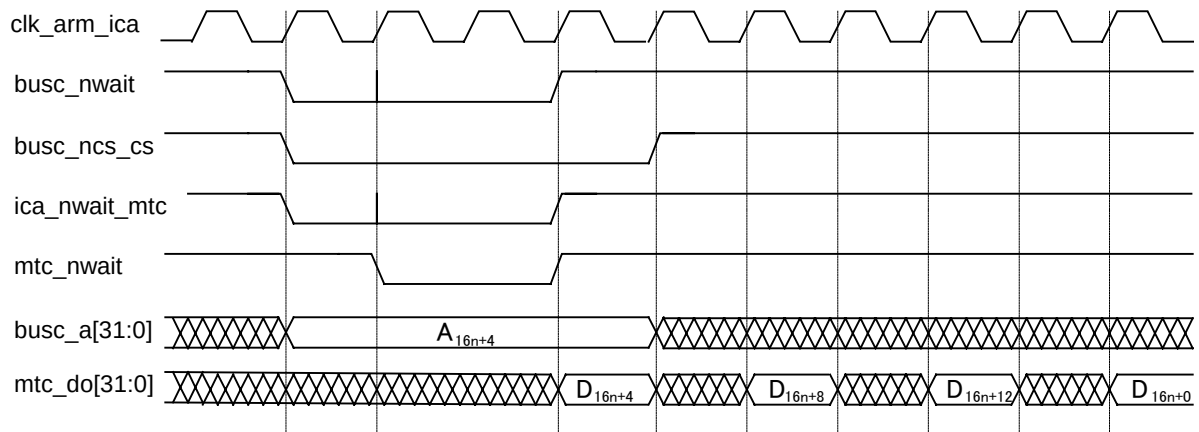


Figure 5 : MTC I/F timing (req4word = 1, Address=16n+4)

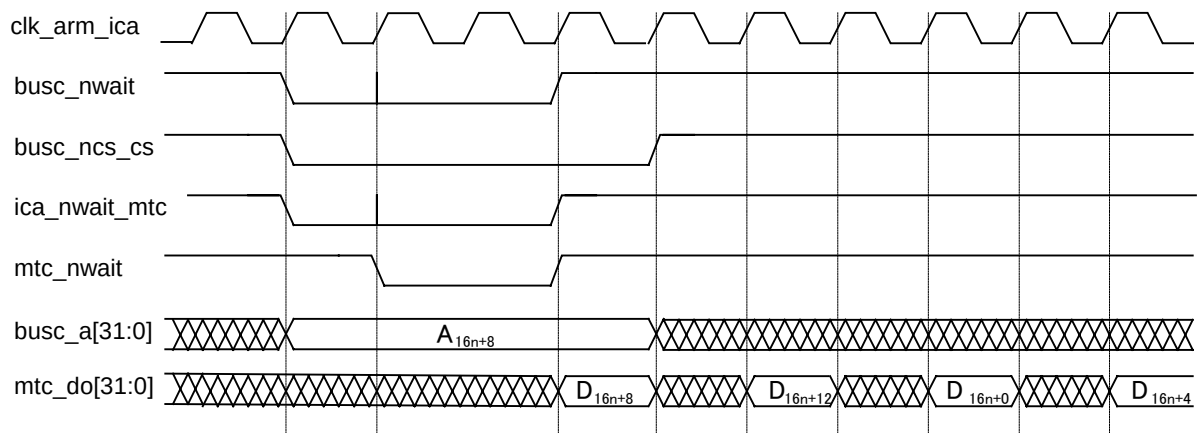


Figure 6 : MTC I/F timing (req4word = 1, Address=16n+8)

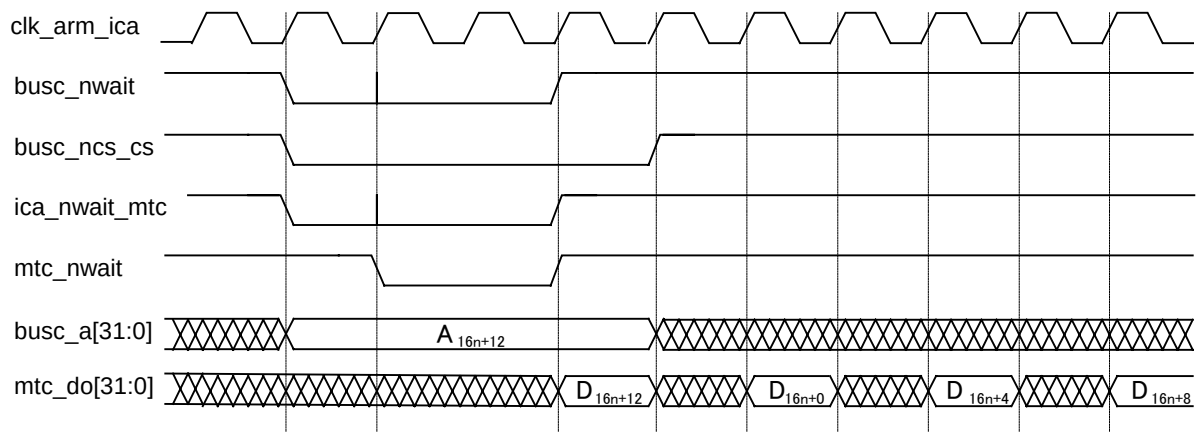


Figure 7 : MTC I/F timing (req4word = 1, Address=16n+12)

5. REGISTERS

Table 1 : ICACHE Register List

Register Name	Offset address	Description
ICAMODE	0x00	ICACHE mode control
ICATEST	0x0F	ICACHE test mode control

ICAMODE (0003:0E00) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	REQ4W	RSV	RSV	RSV	RSV	RSV	RSV	ICACLR	ICAENB
R/W	-	-	-	-	-	-	-	R/W	-	-	-	-	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	0	-	-	-	-	-	-	0	0

Bits	Register Name	Description
0	ICAENB	1: ICACHE enable, 0: ICACHE disable (bypass)
1	ICACLR	1: ICACHE Clear start, 0: ICACHE Clear done
8	REQ4W	1: 4word access request, 0: single word access request

ICACLRビットを1にしたときに、キャッシュのクリアが開始されます。クリア動作中にこのビットを読むと1を返し、クリア完了後に0に戻ります。また、キャッシュクリア動作中にICAENBやREQ4Wが1の場合には、キャッシュのクリア完了後にICACHE動作を開始します。

外部メモリに置いたプログラムが更新されたときは、ICACLRビットを0にする必要があります。

REQ4WはICAENBが1のときのみセットが可能です。

REQ4Wが0のときは、キャッシュとMTCの間でARMからの要求データのみの受け渡しが行われます。

ICATEST (0003:0E00) Offset : 0x7E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ICATST	TAGRD	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV
R/W	R/W	R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Default	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Bits	Register Name	Description
15	ICATST	1: ICACHE Test Mode enable. Invert nOPC signal to cache Data.
14	TAGRD	1: Read Tag Data. Tag RAM Q[44]&[39:22]&[12:0] appear on data bus.

ICATSTを1にするとRead Onlyのデータキャッシュとなります。CACHE RAMに任意のデータを読み書きできるようになります。

TAGRDを1にするとデータリード時にTAG RAMのデータが出力されます。TAGRDはICATSTが1のときのみセットが可能です。

6. Interface

Table 2 : ICACHE Top-level I/O List

Pin Name	Bits	I/O	Fr/To	Description
Functional clocking / common signals				
rst_n	1	I	ext	Power on reset..
clk_arm	1	I	clkc	ARM clock.
clk_tag	1	I	clkc	ARM clock for Tag RAM.
BUSC interface signals (common for External Memory/ICA Reg.)				
busc_a	32	I	busc	Address bus (Ext Mem : 32bit / ICA Reg : LSB 6bit)
busc_do	32	I	busc	Data output bus. (Ext Mem : 32bit / ICA Reg : LSB 16bit)
busc_nrw	1	I	busc	Read/write direction from ARM via BUSC. 0:Read / 1:write.
BUSC interface signals (For External Memory)				
busc_nds	4	I	busc	Data strobe. Active low.
busc_ncs_cs	1	I	busc	Memory Traffic Controller chip select. Active low.
busc_nwait	1	I	busc	Active low wait from BUSC to MTC.
busc_nopc	1	I	busc	'not op-code fetch' signal from ARM via BUSC
ica_do_busc	32	O	busc	Data output bus from MTC to BUSC.
ica_nwait_busc	1	O	busc	Active low wait from MTC to BUSC.
BUSC interface signals (For ICA Reg.)				
busc_ncs_ica	1	I	busc	Chip select. Active low.
ica_do_reg	16	O	busc	ICACHE register data output to BUSC.
MTC interface signals				
mtc_do	32	I	mtc	Data output bus from MTC to ICA.
mtc_nwait	1	I	mtc	Active low wait from MTC to ICA.
mtc_sdssel	1	I	mtc	SDRAM select signal from MTC to ICA. Active high.
ica_a	32	O	mtc	Address bus.
ica_do_mtc	32	O	mtc	Data output bus.
ica_nds	4	O	mtc	Data strobe. Active low.
ica_ncs_cs	1	O	mtc	Memory Traffic Controller chip select. Active low.
ica_nrw	1	O	mtc	Read/write direction from ARM via BUSC. 0:Read / 1:write.
ica_nwait_mtc	1	O	mtc	Active low wait from BUSC to MTC.
lca_req4word	1	O	mtc	1: 4word access request, 0: single word access request
TEST				
tst_scan	1	I	tst	scan test mode enable signal